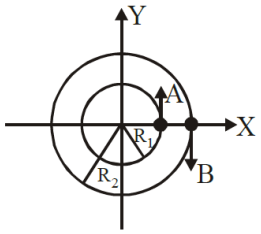
RACE
Regular Analysis Through Continuous Exercise

Circular Motion : RACE-01

1. A particle is moving along a circular path with a constant speed of 10 ms^{-1}. What is the magnitude of the change is velocity of the particle, when it moves through an angle of 60° around the centre of the circle?

(A) zero (B) 10 m/s (C) 10^3 m/s (D) 10^2 m/s

2. Two particles A, B are moving on two concentric circles of radii R_1 and R_2 with equal angular speed ω. At t = 0, their positions and direction of motion are shown in the figure :



The relative velocity $\vec{v}_A - \vec{v}_B$ at $t = \frac{\pi}{2\omega}$ is given by;

(A) $-\omega(R_1 + R_2)\hat{\imath}$ (B) $\omega(R_1 + R_2)\hat{\imath}$ (C) $\omega(R_1 - R_2)\hat{\imath}$ (D) $\omega(R_2 - R_1)\hat{\imath}$

3. A clock has a continuously moving second's hand of 0.1 m length. The average acceleration of the tip of the hand (in units of ms^{-2}) is of the order of :

(A) 10^{-3} ms^{-2} (B) 10^{-2} ms^{-2} (C) 10^{-4} ms^{-2} (D) 10^{-1} ms^{-2}

4. For a particle in uniform circular motion, the acceleration $\vec{a}$ at any point $P(R,\theta)$ on the circular path of radius R is (when θ is measured from the positive x –axis and v is uniform speed) :

(A) $-\frac{v^2}{R}\sin\theta\,\hat{\imath} + \frac{v^2}{R}\cos\theta\,\hat{\jmath}$ (B) $-\frac{v^2}{R}\cos\theta\,\hat{\imath} + \frac{v^2}{R}\sin\theta\,\hat{\jmath}$

(C) $-\frac{v^2}{R}\cos\theta\,\hat{\imath} - \frac{v^2}{R}\sin\theta\,\hat{\jmath}$ (D) $-\frac{v^2}{R}\hat{\imath} + \frac{v^2}{R}\hat{\jmath}$

5. A ball is spun with angular acceleration $\alpha = 6t^2 - 2t$ where t is in second and α is in $rads^{-2}$. At t = 0, the ball has angular velocity of 10 $rads^{-1}$ and angular position of 4 rad. The most appropriate expression for the angular position of the ball is:

(A) $\frac{3}{2}t^4 - t^2 + 10t$ (B) $\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$ (C) $\frac{2t^4}{3} - \frac{t^3}{6} + 10t + 12$ (D) $2t^4 - \frac{t^3}{2} + 5t + 4$

6. An object moves at a constant speed along a circular path in a horizontal plane with centre at the origin. When the object is at x = +2m, its velocity is $-4\hat{\jmath}m/s$. The object's velocity (v) and acceleration (a) at x = –2 m will be

(A) $v = 4\hat{\imath}m/s, a = 8\hat{\jmath}m/s^2$ (B) $v = 4\hat{\jmath}m/s, a = 8\hat{\imath}m/s^2$

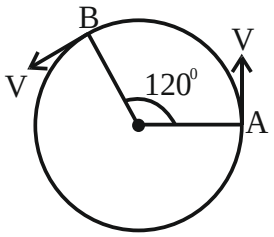
(C) $v = -4\hat{\jmath}m/s, a = 8\hat{\imath}m/s^2$ (D) $v = -4\hat{\imath}m/s, a = -8\hat{\jmath}m/s^2$

7. A body is moving with constant speed, in a circle of radius 10 m. The body completes one revolution in 4s. At the end of 3rd second, the displacement of body (in m) from its starting point is:

(A) 30 (B) 15π (C) 5π (D) $10\sqrt{2}$

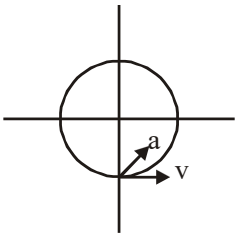
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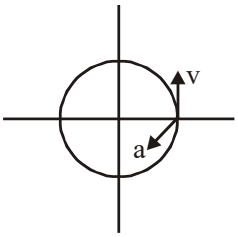
8. As shown in the figure, a particle is moving with constant speed $\pi\ m/s$. Considering its motion from A to B, the magnitude of the average velocity is:

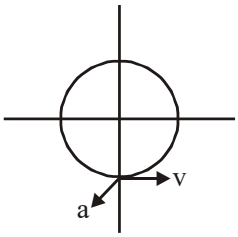


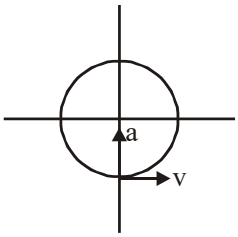
(A) $\pi\ m/s$ (B) $\sqrt{3}\ m/s$ (C) $2\sqrt{3}m/s$ (D) $1.5\sqrt{3}m/s$

9. A particle is undergoing uniformly accelerated circular motion with angular retardation $\pi\ rad/s^2$. If the angular velocity of the particle at t = 0 is $2\pi\ rad/s$, the velocity and acceleration vectors of the body at $t = 1\ s$ are best represented by

(A) 

(B) 

(C) 

(D) 

10. A particle moves with constant angular velocity in a circle of radius $10cm$. If the direction of its velocity reverses after every 100ms then the centripetal acceleration is in ms^{-2}.

(A) 1600 (B) 100 (C) 400 (D) none of these

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	A	C	B	B	D	D	B	B

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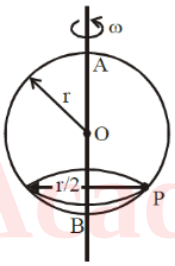
RACE
Regular Analysis Through Continuous Exercise

Circular Motion : RACE-02

1. A body is projected at $t = 0$ with a velocity 10 ms^{-1} at an angle of 60° with the horizontal. The radius of curvature of its trajectory at $t = 1s$ is R. Neglecting air resistance and taking acceleration due to gravity $g = 10\ ms^{-2}$, the value of R is :

(A) 2.5 m (B) 10.3 m (C) 2.8 m (D) 5.1 m

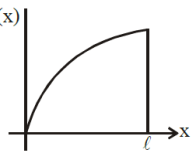
2. A smooth wire of length $2\pi r$ is bent into a circle and kept in a vertical plane. A bead can slide smoothly on the wire. When the circle is rotating with angular speed ω about the vertical diameter AB, as shown in figure, the bead is at rest with respect to the circular ring at position P as shown. Then the value of ω^2 is equal to :

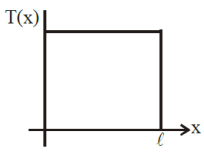


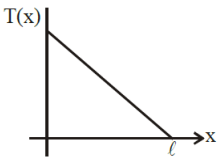
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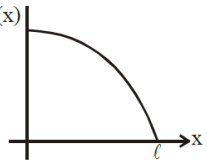
(A) $(g\sqrt{3})/r$ (B) $\frac{\sqrt{3}g}{2r}$ (C) $2g/r$ (D) $2g/(r\sqrt{3})$

3. A uniform rod of length ℓ is being rotated in a horizontal plane with a constant angular speed about an axis passing through one of its ends. If the tension generated in the rod due to rotation is $T(x)$ at a distance x from the axis, then which of the following graphs depicts it most closely?

(A) 

(B) 

(C) 

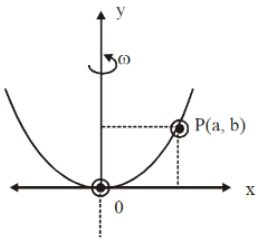
(D) 

4. A spring mass system (mass m, spring constant k and natural length l) rest in equilibrium on a horizontal disc. The free end of the spring is fixed at the centre of the disc. If the disc together with spring mass system, rotates about it's axis with an angular velocity ω, ($k >> m\omega^2$) the relative change in the length of the spring is best given by the option :

(A) $\frac{2m\omega^2}{k}$ (B) $\frac{m\omega^2}{3k}$ (C) $\sqrt{\frac{2}{3}\left(\frac{m\omega^2}{k}\right)}$ (D) $\frac{m\omega^2}{k}$

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5. A bead of mass m stays at point $P(a, b)$ on a wire bent in the shape of a parabola $y = 4Cx^2$ and rotating with angular speed ω (see figure).The value of ω is (neglect friction) :



(A) $\sqrt{\frac{2gC}{ab}}$ (B) $2\sqrt{2gC}$ (C) $\sqrt{\frac{2g}{C}}$ (D) $2\sqrt{gC}$

6. A particle is moving with uniform speed along the circumference of a circle of radius R under the action of a central fictitious force F which is inversely proportional to R^3. Its time period of revolution will be given by :

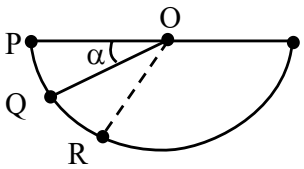
(A) $T \propto R^2$ (B) $T \propto R^{\frac{3}{2}}$ (C) $T \propto R^{\frac{5}{2}}$ (D) $T \propto R^{\frac{4}{3}}$

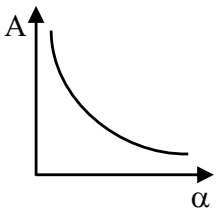
7. A disc with a flat small bottom beaker placed on it at a distance R from its center is revolving about an axis passing through the center and perpendicular to its plane with an angular velocity ω. The coefficient of static friction between the bottom of the beaker and the surface of the disc is μ. The beaker will revolve with the disc if :

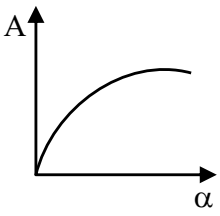
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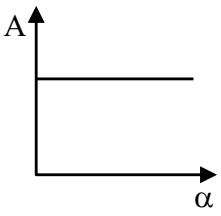
(A) $R \leq \frac{\mu g}{2\omega^2}$ (B) $R \leq \frac{\mu g}{\omega^2}$ (C) $R \geq \frac{\mu g}{2\omega^2}$ (D) $R \geq \frac{\mu g}{\omega^2}$

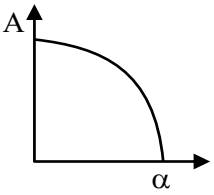
8. A ball is released from rest from point P of a smooth semi-spherical vessel as shown in figure. The ratio of the centripetal force and normal reaction on the ball at point Q is A while angular position of point Q is α with respect to point P. Which of the following graphs represent the correct relation between A and α when ball goes from Q to R ?



(A) 

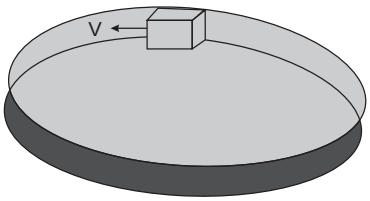
(B) 

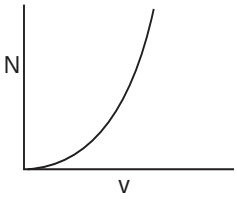
(C) 

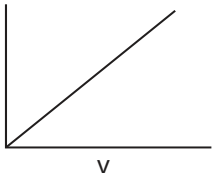
(D) 

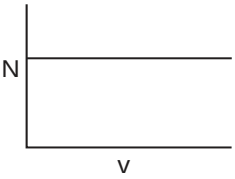
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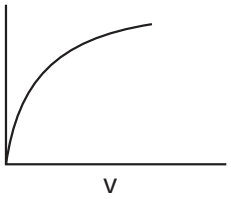
9. A smooth circular groove has a smooth vertical wall as shown in figure. A block of mass m moves against the wall with a speed v. Which of the following curve represents the correct relation between the normal reaction on the block by the wall (N) and speed of the block (v) ?



(A) 

(B) 

(C) 

(D) 

10. A curved in a level road has a radius $75m$. The maximum speed of a car turning this curved road can be 30 m/s without skidding. If radius of curved road is changed to 48 m and the coefficient of friction between the tyres and the road remains same, then maximum allowed speed would be_____ m/s.

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	
Answer	C	D	D	D	B	A	B	C	A	24 m/s	

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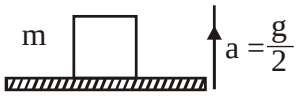
RACE
Regular Analysis Through Continuous Exercise

Work, Energy and Power : RACE-01

1. When a rubber-band is stretched by a distance x, it exerts a restoring force of magnitude $F = ax + bx^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is:-

(A) $\frac{aL^2 + bL^3}{2}$ (B) $\frac{aL^2}{2} + \frac{bL^3}{3}$ (C) $aL^2 + bL^3$ (D) $\frac{1}{2}\left(\frac{aL^2}{2} + \frac{bL^3}{3}\right)$

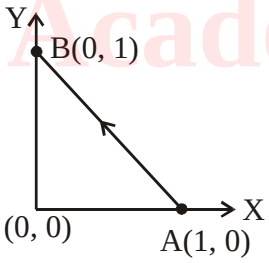
2. A block of mass m is kept on a platform which starts from rest with constant acceleration $g/2$ upward, as shown in fig. Work done *bv* normal reaction on block in time t is :



(A) 0 (B) $\frac{3mg^2t^2}{8}$ (C) $-\frac{mg^2t^2}{8}$ (D) $\frac{mg^2t^2}{8}$

3. Consider a force $\vec{F} = -x\hat{i} + y\hat{j}$. The work done by this force in moving a particle from point A $(1, 0)$ to $B(0, 1)$ along the line segment is :
(all quantities are in SI units)

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(A) $\frac{3}{2}$ (B) 1 (C) 2 (D) $\frac{1}{2}$

4. A person pushes a box on a rough horizontal platform surface. He applies a force of 200 N over a distance of 15 m. Thereafter, he gets progressively tired and his applied force reduces linearly with distance of 100 N. The total distance through which the box has been moved is 30 m. What is the work done by the person during the total movement of the box ?

(A) 5690 J (B) 5250 J (C) 3280 J (D) 2780 J

5. A body of mass 0.5 kg travels on straight line path with velocity $v = (3x^2 + 4) m/s$. The net work done by the force during its displacement from $x = 0$ to $x = 2$ m is :

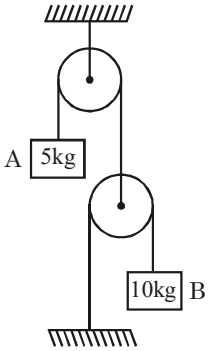
(A) 64 J (B) 60 J (C) 120 J (D) 128 J

6. A force $F = (5 + 3y^2)$ acts on a particle in the y-direction, where F is newton and y is in meter. The work done by the force during a displacement from $y = 2m$ to $y = 5m$ is _____ j.

7. A small particle moves to position $5\hat{i} - 2\hat{j} + \hat{k}$ from its initial position $2\hat{i} + 3\hat{j} - 4\hat{k}$ under the action of force $5\hat{i} + 2\hat{j} + 7\hat{k}$ N. The value of work done will be _____ J.

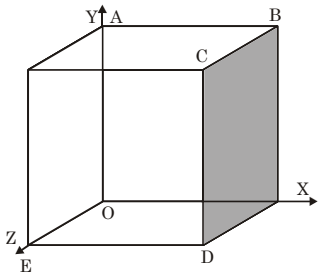
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8. A system of 5*kg* and 10*kg* blocks is shown in figure and strings and pulley are ideal. If 10*kg* block goes down by 10*m* then find work done by tension on 5*kg* block.



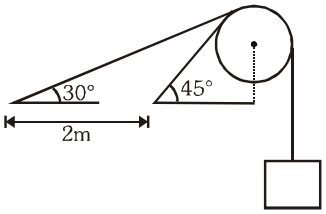
(A) $\frac{500}{3}$J (B) $\frac{-500}{3}$J (C) $\frac{-1000}{3}$J (D) $\frac{1000}{3}$J

9. A particle moves under the influence of the force $F = yi + zj + xk$ *N* along the edges of a unit cube (edge length = 1*m*) on the path *OA-AB-BC-CD-DE-EO*. Then the :-



(A) Total work done is zero (B) Total work done is 1 joule
(C) Total work done is 3 joule (D) Force *F* is conservative

10. A person lifts a 0.1 *kg* crate hanging over a pulley by walking horizontally, as shown in figure. As the person walks 2*m* the angle of the rope to the horizontal changes from 45° to 30°. How much work (in Joules) does the person do if the crate rises at constant speed.



(A) $\frac{\sqrt{2}(\sqrt{3}+1)}{(\sqrt{2}+1)}$ (B) $\frac{\sqrt{3}(\sqrt{3}+1)}{(\sqrt{2}+1)}$ (C) $\frac{\sqrt{2}(\sqrt{3}-1)}{(\sqrt{2}+1)}$ (D) $\frac{2\sqrt{2}(\sqrt{3}+1)}{(\sqrt{2}+1)}$

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	
Answer	B	B	B	B	B	132	40	D	B	A	

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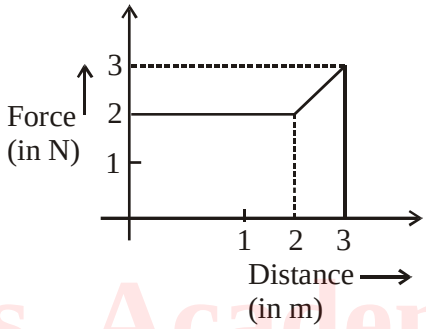
RACE
Regular Analysis Through Continuous Exercise

Work, Energy and Power : RACE-02

1. A time dependent force $F = 6t$ acts on a particle of mass 1 *kg*. If the particle starts from rest, the work done by the force during the first 1 *sec*. will be :

(A) 4.5 *J* (B) 22 *J* (C) 9 *J* (D) 18 *J*

2. A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3*m* is :



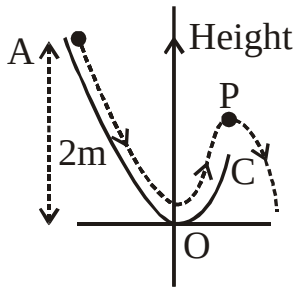
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(A) 6.5 J (B) 2.5 J (C) 4 J (D) 5 J

3. A body of mass 1 *kg* falls freely from a height of 100 *m* on a platform of mass 3 *kg* which is mounted on a spring having spring constant $k = 1.25 \times 10^6$ N/m. The body sticks to the platform and the spring's maximum compression is found to be *x*. Given that $g = 10$ ms^{-2}, the value of *x* will be close to:

(A) 4 *cm* (B) 8 *cm* (C) 80 *cm* (D) 40 *cm*

4. A particle ($m = l$ *kg*) slides down a frictionless track (*AOC*) starting from rest at a point *A* (height 2 *m*). After reaching *C*, the particle continues to move freely in air as a projectile. When it reaching its highest point *P* (height 1 *m*), the kinetic energy of the particle (in *J*) is : (Figure drawn is schematic and not to scale; take $g = 10$ ms^{-2})_____.



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5. Arrange the four graphs in descending order of total work done; where W_1, W_2, W_3 and W_4 are the work done corresponding to figure *a*, *b*, *c* and *d* respectively.

Figure-a

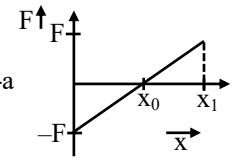


Figure-b

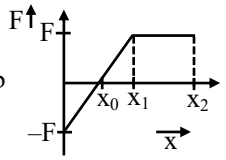


Figure-c

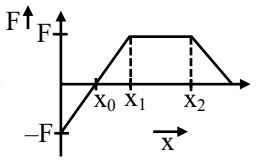
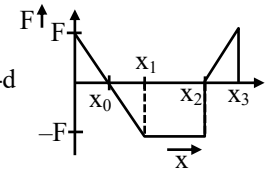


Figure-d

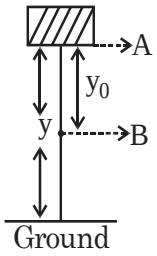


(A) $W_3 > W_2 > W_1 > W_4$ (B) $W_3 > W_2 > W_4 > W_1$

(C) $W_2 > W_3 > W_4 > W_1$ (D) $W_2 > W_3 > W_1 > W_4$

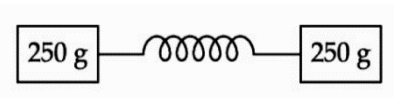
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6. In the given figure, the block of mass *m* is dropped from the point '*A*'. The expression for kinetic energy of block when it reaches point '*B*' is :



(A) $\frac{1}{2}mg\,y_0^2$ (B) $\frac{1}{2}mg\,y^2$ (C) $mg(y - y_0)$ (D) mgy_0

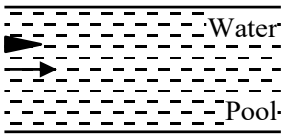
7. As per the given figure, two blocks each of mass $250g$ are connected to a spring of spring constant $2Nm^{-1}$. If both are given velocity v in opposite directions, then maximum elongation of the spring is :



(A) $\frac{v}{2\sqrt{2}}$ (B) $\frac{v}{2}$ (C) $\frac{v}{4}$ (D) $\frac{v}{\sqrt{2}}$

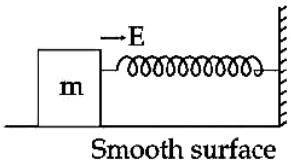
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8. A bullet of mass 200 g having initial kinetic energy 90 J is shot inside a long swimming pool as shown in the figure. If it's kinetic energy reduces to 40 J within 1s, the minimum length of the pool, the bullet has a to travel so that it completely comes to rest is



(A) 45 m (B) 90 m (C) 125 m (D) 25 m

9. A block of mass 'm' (as shown in figure) moving with kinetic energy E compresses a spring through a distance 25 cm when, its speed is halved. The value of spring constant of used spring will be nE Nm^{-1} for n = _____.



10. A person trying to lose weight by burning fat lifts a mass of 10 kg upto a height of 1 m 1000 times. Assume that the potential energy lost each time he lowers the mass is dissipated. How much fat will he use up considering the work done only when the weight is lifted up? Fat supplies 3.8×10^7 J of energy per kg which is converted to mechanical energy with a 20% efficiency rate. Take $g = 9.8$ ms^{-2} :-

(A) 12.89×10^{-3} kg (B) 2.45×10^{-3} kg (C) 6.45×10^{-3} kg (D) 9.89×10^{-3} kg

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	
Answer	A	A	A	10	A	D	B	A	24	A	

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RACE
Regular Analysis Through Continuous Exercise

Work, Energy and Power : RACE-03

1. A spring of unstretched length l has a mass m with one end fixed to a rigid support. Assuming spring to be made of a uniform wire, the kinetic energy possessed by it if its free end is pulled with uniform velocity v is :-

(A) $\frac{1}{6}mv^2$ (B) mv^2 (C) $\frac{1}{3}mv^2$ (D) $\frac{1}{2}mv^2$

2. A uniform cable of mass 'M' and length 'L' is placed on a horizontal surface such that its $\left(\frac{1}{n}\right)^{th}$ part is hanging below the edge of the surface. To lift the hanging part of the cable upto the surface, the work done should be :

(A) $\frac{MgL}{n^2}$ (B) $\frac{MgL}{2n^2}$ (C) $\frac{2MgL}{n^2}$ (D) nMgL

3. Water fall from a 40 m high dam at the rate of $9 \times 10^4 kg$ per hour. Fifty percentage of gravitational potential energy can be converted into electrical energy. Using this hydroelectric energy number of $100W$ lamps, that can be lit, is:

(Take g = 10 ms^{-2})

(A) 25 (B) 50 (C) 100 (D) 18

4. Two particles of the same mass m are moving in circular orbits because of force, given by $F(r)=\frac{-16}{r}-r^3$. The first particle is at a distance r = 1, and the second, at r = 4. The best estimate for the ratio of kinetic energies of the first and the second particle is closest to :-

(A) 6×10^{-2} (B) 10^{-1} (C) 3×10^{-3} (D) 6×10^{2}

5. Two bodies are having kinetic energies in the ratio 16 : 9. If they have same linear momentum, the ratio of their masses respectively is :

(A) 4 : 3 (B) 3 : 4 (C) 16 : 9 (D) 9 : 16

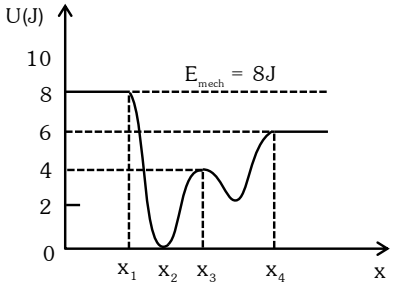
6. If the potential energy between two molecules is given by $U=\frac{A}{r^6}+\frac{B}{r^{12}}$, then at equilibrium, separation between molecules, and the potential energy are :

(A) $\left(\frac{B}{A}\right)^{1/6},0$ (B) $\left(\frac{B}{2A}\right)^{1/6},-\frac{A^2}{2B}$ (C) $\left(\frac{2B}{A}\right)^{1/6},-\frac{A^2}{4B}$ (D) $\left(\frac{2B}{A}\right)^{1/6},-\frac{A^2}{2B}$

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7. Given below is the plot of a potential energy function $U(x)$ for a system, in which a particle is in one dimensional motion, while a conservative force $F(x)$ acts on it. Suppose that $E_{mech} = 8\,J$, the incorrect statement for this system is :



[where K.E. = kinetic energy]

(A) at $x > x_4$, *K.E.* is constant throughout the region.
(B) at $x < x_1$, *K.E.* is smallest and the particle is moving at the slowest speed.
(C) at $x = x_2$, *K.E.* is greatest and the particle is moving at the fastest speed.
(D) at $x = x_3$, *K.E.* = 4 *J*.

8. A particle is moving in a circular path of radius a under the action of an attractive potential $U = -\frac{k}{2r^2}$. Its total energy is :-

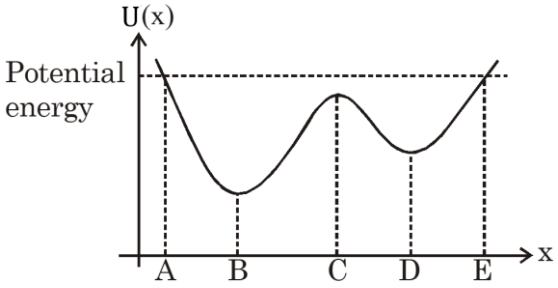
(A) $\frac{k}{2a^2}$ (B) Zero (C) $-\frac{3}{2}\frac{k}{a^2}$ (D) $-\frac{k}{4a^2}$

9. A particle is moving in a circle of radius r under the action of a force $F = \alpha r^2$ which is directed towards centre of the circle. Total mechanical energy (kinetic energy + potential energy) of the particle is (take potential energy = 0 for $r = 0$) :-

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(A) $\frac{4}{3}\alpha r^3$ (B) αr^3 (C) $\frac{5}{6}\alpha r^3$ (D) $\frac{1}{2}\alpha r^3$

10. A particle moves in one dimension in a conservative force field. The potential energy is depicted in the graph below. If the particle starts to move from rest from the point A, then:



(A) The speed is zero at the points A and E
(B) The acceleration vanishes at the points A,B,C,D,E
(C) The acceleration vanishes at the points B,C,D
(D) The speed is maximum at the point D

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	B	B	A	D	C	B	B	C	AC

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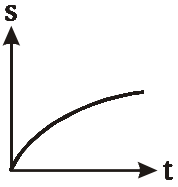
RACE
Regular Analysis Through Continuous Exercise

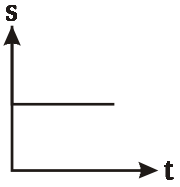
Work, Energy and Power : RACE-04

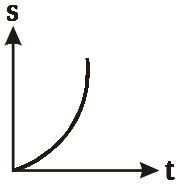
1. An elevator in a building can carry a maximum of 10 persons, with the average mass of each person being 68 kg. The mass of the elevator itself is 920 kg and it moves with a constant speed 3 m/s. The frictional force opposing the motion is 6000 N. If the elevator is moving up with its full capacity, the power delivered by the motor to the elevator (g = 10 m/s^2) must be at least :

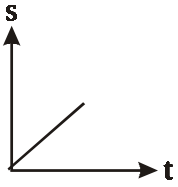
(A) 56300 W (B) 48000 W (C) 66000 W (D) 62360 W

2. A particle is moving unidirectionally on a horizontal plane under the action of a constant power supplying energy source. The displacement (s) - time (t) graph that describes the motion of the particle is (graphs are drawn schematically and are not to scale) :

(A) 

(B) 

(C) 

(D) 

3. A body of mass $2kg$ is driven by an engine delivering a constant power $1J/s$. The body starts from rest and moves in a straight line. After 9 seconds, the body has moved a distance (in m) _____.

4. A pendulum bob has a speed of 3 m/s at its lowest position. The pendulum is 50 cm long. The speed of bob, when the length makes an angle of 60° to the vertical will be (g = 10 m/s^2) _____ m/s.

5. An automobile of mass 'm' accelerates starting from origin and initially at rest, while the engine supplies constant power P. The position is given as a function of time by :

(A) $\left(\frac{9P}{8m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$ (B) $\left(\frac{8m}{9P}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$ (C) $\left(\frac{9m}{8P}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$ (D) $\left(\frac{8P}{9m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$

6. A stone tide to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of change in its velocity, as it reaches a position where the string is horizontal, is $\sqrt{x(u^2 - gL)}$. The value of x is

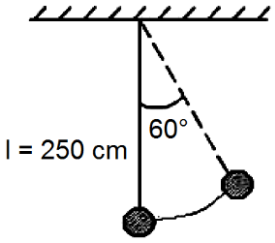
(A) 3 (B) 2 (C) 1 (D) 5

7. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration is varying with time t as $a = k^2rt^2$. where k is a constant. The power delivered to the particle by the force acting on it is given as

(A) zero (B) $mk^2r^2t^2$ (C) mk^2r^2t (D) mk^2rt

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8. A pendulum is suspended by a string of length 250 cm. The mass of the bob of the pendulum is 200 g. The bob is pulled aside until the string is at 60° with vertical as shown in the figure. After releasing the bob. the maximum velocity attained by the bob will be _____ ms^{-1}. (if $g = 10\ m/s^2$)



9. A body of mass 2 kg is initially at rest. It starts moving unidirectionally under the influence of a source of constant power P. Its displacement in $4s$ is $\frac{1}{3}\alpha^2\sqrt{P}\,m$. The value of α will be .....

10. The ratio of powers of two motors is $\frac{3\sqrt{x}}{\sqrt{x}+1}$, that are capable of raising 300 kg water in 5 minutes and 50 kg water in 2 minutes respectively from a well of 100 m deep. The value of x will be

(A) 2 (B) 4 (C) 2.4 (D) 16

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	
Answer	C	C	18.00	2	D	B	C	5	4	D	

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